A Sleep Monitoring Belt–Based Sleep Staging Method via EEG-Guided Label Transfer and Deep Learning

Jian Zhan[#], Xinran Lin[#], Shangle Ye[#], Heng Wu, Jing Lin*, Maojin Liang*, Chunhua He*

Abstract—Polysomnography (PSG) remains the reference standard for sleep staging, but its cost, operational complexity, and laboratory dependence limit long-term home monitoring. Non-wearable sleep monitoring belts are more comfortable and scalable, yet their development is constrained by scarce medical-grade labels. This study proposes an EEG-guided label-transfer framework for belt-based four-stage sleep staging. A source-domain model was trained on single-channel Sleep-EDF EEG (Fpz-Cz) and evaluated with subject-wise splits to reduce epoch-level leakage. Synchronized overnight EEG and inertial measurement unit (IMU) data were then collected from 40 healthy participants. The EEG model generated temporally aligned labels, which were transferred to IMU segments through hardware-level synchronization. Under subject-wise Sleep-EDF evaluation, the proposed model achieved an accuracy of 0.8662, a macro-F1 of 0.8410, and a Cohen's kappa of 0.7904. Additional wake-window analyses showed accuracies of 0.9338 and 0.8714. In the target domain, the IMU model trained with EEG-transferred labels achieved 83.56% accuracy, 79.69% macro-F1, and a Cohen's kappa of 0.7498 in a feasibility evaluation. These results support synchronized EEG-guided label transfer as a practical route for developing non-wearable sleep staging models, while broader clinical validation remains necessary.

***Index Terms*—Sleep staging, sleep monitoring belt, label transfer, deep learning, inertial measurement unit (IMU).**

I. Introduction

Sleep disorders have emerged as a significant public health concern [1]. According to relevant epidemiological studies [2-4], approximately 27% of adults worldwide experience insomnia symptoms [5-7], with chronic insomnia affecting 10%–15% of the general population [8-9]. These prevalence rates are notably higher among older adults, women, and individuals with chronic medical conditions [10-12]. Persistent deterioration in sleep quality substantially impairs daytime functioning and quality of life [13-15], and is strongly associated with a range of chronic diseases, including cardiovascular disorders [16-17], depression [18-19], cognitive decline [20-21], and Alzheimer's disease [22-23]. Accurate assessment and effective intervention for sleep disorders hinge on the evaluation of sleep duration and the fine-grained analysis of sleep architecture [24-25]. Nocturnal sleep is characterized by distinct stages that cycle periodically throughout the night, which are commonly categorized in macro-architecture analysis as Wake (W), Light Sleep (N1 + N2), Deep Sleep (N3), and Rapid Eye Movement (REM) sleep [25]. The relative proportions and transition dynamics among these stages reflect the regulatory and restorative capacity of the central nervous system [26-27]. Consequently, precise identification of each sleep stage holds substantial clinical and scientific value [28].

Polysomnography (PSG) is internationally recognized as the gold standard for sleep staging [29-30]. Multimodal physiological signals are simultaneously recorded, including electroencephalography (EEG), electrooculography (EOG), electromyography (EMG), electrocardiography (ECG), respiratory airflow, blood oxygen saturation, and body position [31-32]. Based on manual scoring by certified technicians in accordance with the American Academy of Sleep Medicine (AASM) guidelines [33-34], these signals are originally classified into five stages (W, N1, N2, N3, and REM). For automated sleep staging tasks, they are commonly regrouped into four classes: Wake, Light Sleep (N1 + N2), Deep Sleep (N3), and REM sleep. Owing to its high temporal resolution and comprehensive physiological correlation [33], PSG plays a critical role in the diagnosis of conditions such as insomnia, obstructive sleep apnea, and narcolepsy [35]. However, significant limitations that hinder broader application are associated with PSG [36-37]. A specialized sleep laboratory and expensive equipment are required, resulting in high operational costs. The necessity of attaching numerous sensors is highly intrusive and uncomfortable for participants [38], often triggering the "first-night effect," which distorts natural sleep architecture [39]. Moreover, PSG provides only a snapshot assessment and is unsuitable for long-term, continuous monitoring in home settings [40-41]. These constraints severely limit its feasibility for large-scale population screening and routine health management.

Various unobtrusive sleep monitoring approaches have been proposed, including photoplethysmography (PPG)-based wristbands [41], millimeter-wave radar [42], and mattress-embedded sensors [43]. Recent wearable sleep-staging studies

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have further extended four-class PSG-referenced classification to clinical populations by integrating cardiac and motion-related signals, such as PPG-derived heart activity and accelerometer-based body movement [44]. However, accurate four-stage sleep staging remains difficult when the sensing modality captures indirect physiological or motion-related signatures. The sleep monitoring belt used in this study was developed in prior work [45] and integrates triaxial acceleration and angular-velocity sensing on the mattress surface. It can record whole-night body movement and posture-related micromotion without attaching sensors to the participant. In earlier use, however, belt-based staging relied mainly on statistical rules or threshold models and achieved approximately 78% accuracy. The main limitation is not the amount of belt data, but the lack of reliable EEG-referenced labels for training high-capacity models. This label scarcity has become a central barrier for contactless home sleep staging systems [47].

To address this barrier, this paper develops an EEG-guided label-transfer method for a non-wearable sleep monitoring belt. First, candidate classifiers were evaluated on single-channel frontal EEG (Fpz-Cz) from the public Sleep-EDF dataset [48]. The source-domain evaluation was performed using subject-wise cross-validation and Cohen's kappa, rather than relying on epoch-wise accuracy alone. Second, overnight EEG and IMU data were recorded synchronously from 40 healthy participants using a portable EEG system [49] and the mattress-based belt [45]. The target-domain EEG recordings were sampled at 100 Hz and normalized at the recording level using robust z-score standardization before epoch segmentation and model input. The trained EEG classifier then generated sleep-stage pseudo-labels for each 30-s epoch. Third, hardware synchronization with an alignment error below 10 ms mapped these labels to the corresponding IMU segments. Finally, lightweight models were trained on the transferred target-domain labels. The resulting IMU model is reported as a feasibility evaluation of the label-transfer pipeline, not as evidence of clinical deployment readiness.

The main contributions of this study are as follows:

(1) We propose an EEG-guided label-transfer paradigm for contactless sleep monitoring. The approach uses synchronized EEG-derived sleep-stage labels to supervise a mattress-based IMU device, thereby addressing the data-rich but label-poor setting common in home sleep monitoring.

(2) We evaluate an IMU-based target model trained with transferred EEG labels. The model uses synchronized acceleration and angular-velocity signals together with IMU-derived heart rate, respiratory rate, and activity features. Its confusion matrix and a representative transferred-label hypnogram are reported to show feasibility and class-specific behavior.

II. Materials and Methods

The overall technical framework is shown in Fig. 1. The pipeline contains two stages: leakage-controlled source-domain modeling on Sleep-EDF and target-domain label transfer for the non-wearable belt. In the source domain, Sleep-EDF provides single-channel frontal EEG signals (Fpz-Cz, 100 Hz) with expert sleep-stage annotations. The original Sleep-EDF labels are mapped to four stages: Wake, Light Sleep (N1+N2), Deep Sleep (N3+N4), and REM. This grouping follows common four-class automatic staging practice, but it also means that N1-specific fragmentation is not separately assessed.

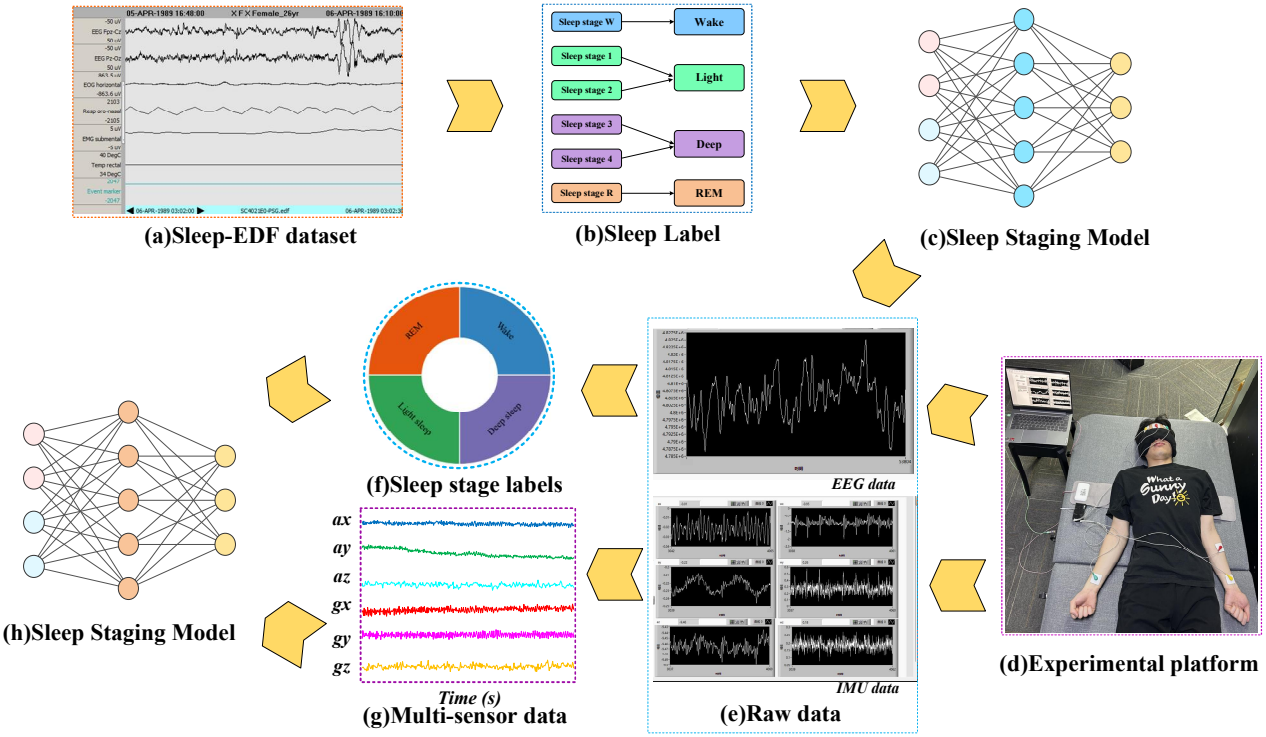


Fig. 1. Overall framework of EEG-guided label transfer for belt-based sleep staging. The source-domain stage trains and validates a Sleep-EDF EEG model, and the target-domain stage transfers synchronized EEG labels to IMU segments for belt-model training.

In the target domain, multimodal data were collected from 40 healthy participants in a simulated home environment. The acquisition platform included a mask-integrated portable EEG system [49] and the non-wearable belt [45]. The portable EEG system recorded target-domain EEG signals at 100 Hz, while the belt continuously recorded triaxial acceleration and angular velocity. Heart rate (HR), respiratory rate (RR), and activity indices were also extracted from the belt signal and used as target-domain inputs.

The two devices were aligned by a hardware trigger with an error below 10 ms. After the EEG model assigned sleep-stage labels to the normalized EEG epochs, the labels were transferred to temporally matched IMU epochs. The resulting paired dataset links each 30-s belt segment with an EEG-guided sleep-stage label. The target-domain model uses the same temporal modeling principle as Fig. 2, but its input features differ from the Sleep-EDF source-domain inputs

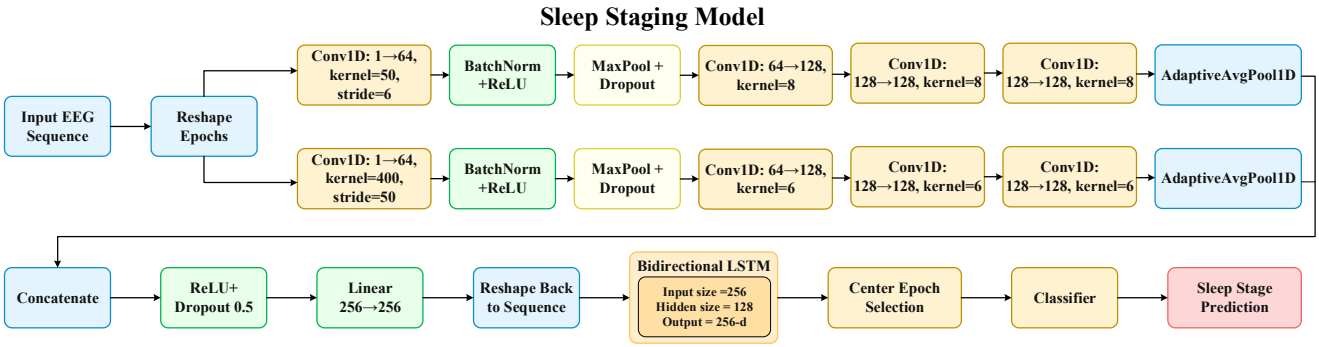


Fig. 2. Deep temporal sleep-stage model structure. For Sleep-EDF, the input consists of preprocessed raw single-channel Fpz-Cz 30-s EEG epochs. For the target-domain IMU model, the input consists of synchronized triaxial acceleration, triaxial angular velocity, and belt-derived HR, RR, and activity features. The bidirectional LSTM models contextual dependencies across neighboring epochs.

A. Data preprocessing

The publicly available expanded Sleep-EDF dataset [48] was used as the source-domain dataset. Only the Fpz-Cz EEG channel was used, and all recordings were segmented into standard 30-s epochs. To avoid leakage, train-test assignment was performed at the subject level, so epochs from the same participant were not shared across training and testing folds.

The sleep-cassette (SC) group and the sleep-telemetry (ST) group from the sleep-edf dataset were selected for model development. The Fpz-Cz channel was chosen because frontal single-lead EEG is compatible with portable sleep monitoring and preserves key sleep-stage features, including slow waves and spindle-related activity. The channel choice is compatible with the target-domain EEG recordings, which were sampled at the same frequency and normalized using the same record-wise robust standardization strategy before model input.

The original Sleep-EDF annotations were mapped to the following four-stage labels:

- Wake (W) → 0
- Light Sleep (N1 + N2) → 1
- Deep Sleep (N3 + N4) → 2
- REM (R) → 3

After label mapping, each recording was divided into non-overlapping 30-s epochs. Only epochs with valid Wake, N1, N2, N3, N4, or REM annotations were retained. Because the Fpz-Cz channel was sampled at 100 Hz, each Sleep-EDF epoch corresponded to a 3000-sample EEG segment. No handcrafted time-domain or frequency-domain feature vector was constructed. The preprocessed Fpz-Cz epoch was used directly as the Sleep-EDF input shown in Fig. 2.

For each subject-wise training fold, signal normalization parameters were estimated from the training recordings and then applied to the held-out recordings. This fold-wise preprocessing avoided information leakage from validation or test subjects.

To prevent information leakage, all data partitions were performed at the subject level, ensuring that subjects in the training, validation, and test sets were mutually exclusive. For wake-stage processing, the commonly used Trimmed setting retains only wake epochs within 30 min before sleep onset and 30 min after final awakening. However, in our subsequent experimental setting, a substantial number of wake epochs may still occur before sleep onset, which better reflects the intended target application scenario. Therefore, we also adopted a Post30 setting in this study, retaining all wake epochs before sleep onset while keeping only the 30-min wake period after final awakening, so as to better match the target experimental environment.

B. Data collection

To enable reliable label transfer from EEG signals to non-wearable sensor signals, a synchronized multimodal data acquisition experiment was designed in this study.

Participant recruitment: Forty healthy adult volunteers were recruited. The inclusion criteria were absence of known sleep disorders, no history of neurological or psychiatric disease, and no recent medication affecting the central nervous system. All participants provided written informed consent, and the study was approved by the institutional ethics committee. This cohort supports feasibility testing of synchronized label transfer, but it did not include patients with sleep disorders. Clinical generalization therefore remains to be evaluated in future patient cohorts.

Experimental equipment: Target-domain EEG signals were recorded using a portable EEG system [49] at a sampling rate of 100 Hz, consistent with the 100-Hz sampling rate of the Sleep-EDF EEG recordings. To mitigate amplitude-scale

discrepancies across recordings, both the target-domain EEG and Sleep-EDF EEG signals were independently normalized at the recording level using robust z-score standardization. Specifically, recording-level measures of central tendency and robust dispersion were used to transform EEG amplitudes into a dimensionless representation prior to 30-s epoch segmentation and model input. This preprocessing strategy harmonized the input scale across datasets; however, it did not reconstruct the Fpz-Cz montage or fully eliminate domain shifts arising from differences in acquisition devices, montages, or electrode placements. Therefore, the transferred labels should be regarded as EEG-guided pseudo-labels rather than independent PSG-based annotations.

Target device: The non-wearable sleep monitoring belt [45] was placed on the mattress under the participant's torso. The belt recorded triaxial acceleration and triaxial angular velocity at 100 Hz without requiring body-worn sensors. HR, RR, and activity features were derived from the belt signal and combined with the six motion channels for target-domain modeling.

The two systems were synchronized by a hardware trigger, achieving an alignment error below 10 ms. Each participant completed a full-night recording of approximately 7-8 h in a quiet, temperature-controlled laboratory environment. The normalized EEG signal was segmented into 30-s epochs and processed by the source-domain model to generate four-class sleep-stage labels. After confidence filtering, these labels were transferred to the temporally aligned IMU epochs.

C. Label transfer and target model training

The trained EEG model generated a posterior probability vector for each synchronized 30-s target-domain EEG epoch. The predicted sleep stage was determined as the class with the highest posterior probability, and the corresponding maximum probability was used as the confidence score of the EEG-derived pseudo-label. To reduce the influence of uncertain pseudo-labels, only epochs with confidence scores of 0.80 or higher were retained for IMU-label model training. The retained EEG-derived labels were subsequently assigned to the temporally aligned IMU epochs based on hardware synchronization. The paired target-domain dataset, D_{target}, was constructed as:

$$D_{target} = \{(x_i^{IMU}, y_i^{EEG})\}_{i=1}^{N} \quad (1)$$

where x_i^{IMU} denotes the i-th 30-second segment of triaxial acceleration and angular velocity signals, and $y_i^{EEG} \in \{0, 1, 2, 3\}$ represents the corresponding sleep stage label transferred from the EEG-derived pseudo-labels.

For the target-domain IMU model, only IMU epochs temporally aligned with the retained EEG-derived pseudo-labels were used for model training and evaluation. The IMU input comprised triaxial acceleration, triaxial angular velocity, and belt-derived physiological and activity features, including HR, RR, and activity. The raw motion channels were first filtered using a sliding-window median filter and then segmented into non-overlapping 30-s epochs synchronized with the EEG epochs. HR, RR, and activity features were extracted from the corresponding 30-s windows and concatenated with the motion representation to form the target-domain input. The extraction of HR and RR followed the same procedure described in our previous study [45]. To prevent information leakage, all continuous IMU input channels were standardized using Z-score normalization, with normalization statistics estimated exclusively from the training set and subsequently applied to the validation and test sets. The IMU model adopted the same temporal modeling backbone as shown in Fig. 2; however, its input representation differed from that of the EEG branch, and the parameters of several modules were accordingly adjusted. The model was supervised using confidence-filtered EEG-derived pseudo-labels.

$$\tilde{x}_{i,c}^{IMU} = \frac{x_{i,c}^{IMU} - \mu_c^{train}}{\sigma_c^{train}} \quad (2)$$

where $x_{i,c}^{IMU}$ denotes the value of channel c in the i-th IMU epoch, and μ_c^{train} and σ_c^{train} denote the mean and standard deviation of that channel computed from the training set. The same normalization parameters were then applied to the validation or test data to avoid information leakage.

The target-domain model used the same temporal modeling backbone shown in Fig. 2, but with a target-specific IMU input representation. The model was trained on temporally aligned IMU epochs paired with EEG-derived pseudo-labels. Performance was assessed using accuracy, macro-F1, recall, Cohen's kappa, and class-level confusion matrices.

D. Model Implementation and Training Protocol

All models were implemented using PyTorch 2.7.1. Both the source-domain EEG model and the target-domain IMU model were trained for four-class sleep staging, including Wake, Light Sleep, Deep Sleep, and REM. For the Sleep-EDF EEG experiments, subject-wise partitioning was adopted to ensure that the training, validation, and test sets contained mutually exclusive subjects.

The source-domain EEG sleep-staging model was optimized using AdamW with an initial learning rate of 5×10^{-4}, a batch size of 32, and a maximum of 200 training epochs. Four-class cross-entropy loss with label smoothing of 0.03 was used as the training objective. Dropout was applied to the CNN encoder and classifier, with dropout probabilities of 0.2 and 0.5, respectively.

For temporal modeling, consecutive 30-s epochs were grouped into sequences of 21 epochs, and the model predicted the sleep stage of the center epoch. The EEG input consisted of raw single-channel Fpz-Cz 30-s epochs from Sleep-EDF. A two-branch CNN encoder with small and large temporal filters was used to extract epoch-level representations, followed by a bidirectional LSTM to capture contextual dependencies across neighboring epochs.

For the target-domain IMU sleep stage model, the input consisted of triaxial acceleration and triaxial angular-velocity signals. In the HR/RR setting, heart rate, respiratory rate, and activity features estimated from the corresponding IMU epochs were appended as additional channels. The IMU model was

optimized using AdamW with an initial learning rate of 1×10^{-3}, a batch size of 64, and a maximum of 200 training epochs. Four-class cross-entropy loss was used, with class weights estimated from the training set and label smoothing of 0.03. Dropout probabilities of 0.35 and 0.4 were used in the convolutional branches and classifier head, respectively.

The target-domain IMU model was supervised using EEG-derived pseudo-labels transferred from the trained single-channel EEG sleep-staging model. All normalization parameters, class weights, and model-selection decisions were estimated or determined using only the training or validation data.The held-out test set was used exclusively for final evaluation.

III. Results and Discussion

A. Model performance comparison on the Sleep-EDF dataset

Source-domain performance was evaluated on Sleep-EDF with subject-wise splitting. This design prevents epochs from the same participant from appearing in both training and testing folds, thereby reducing the risk of subject-level information leakage. The four-class labels were Wake, Light Sleep, Deep Sleep, and REM.

TABLE I
EXPERIMENTAL RESULTS UNDER SUBJECT-WISE SPLITTING

Methods	Datasets	Acc	MF1	Kappa	Recall
TinySleepNet	(SC+ST)	0.8633	0.8433	0.7849	0.8284
AttnSleep	SC	0.8656	0.8379	0.7930	0.8292
ours	(SC+ST)	0.8662	0.8410	0.7904	0.8283

As shown in Table I, our model achieved 0.8662 accuracy, 0.8410 macro-F1, 0.7904 Cohen's kappa, and 0.8283 recall under subject-wise evaluation. Its performance was comparable to TinySleepNet [50] and AttnSleep [46] under the listed Sleep-EDF settings. They provide a more reliable basis for selecting the model as the EEG label generator.

TABLE II
SLEEP-EDF 5-FOLD SUBJECT-WISE CROSS-VALIDATION UNDER TWO WAKE-WINDOW STRATEGIES

Methods	Acc ± SD	MF1 ± SD	Kappa ± SD	Recall ± SD
Post30	0.9338 ± 0.0069	0.8682 ± 0.0060	0.8754 ± 0.0062	0.8666 ± 0.0162
trimmed	0.8714 ± 0.0071	0.8466 ± 0.0082	0.8015 ± 0.0108	0.8350 ± 0.0147

Table II reports the Sleep-EDF 5-fold subject-wise cross-validation results under two wake-window strategies, and Figs. 3 and 4 show the corresponding class-level confusion matrices. This analysis was designed to evaluate how the source-domain model behaved under wake-retention settings that differ in their consistency with the target-domain experimental protocol. In the Post30 setting, all pre-sleep wake epochs were retained, whereas only 30 min of post-sleep wake epochs were kept. This strategy was used to match the actual experimental setting in the target domain, where the complete wake period before sleep onset was preserved but the post-sleep wake recording was limited. Under this setting, the model achieved an accuracy of 0.9338 ± 0.0069 and a kappa of 0.8754 ± 0.0062 (Table II; Fig. 3). In the trimmed setting, both pre-sleep and post-sleep wake periods were limited to 30 min, yielding an accuracy of 0.8714 ± 0.0071 and a kappa of 0.8015 ± 0.0108 (Table II; Fig. 4).

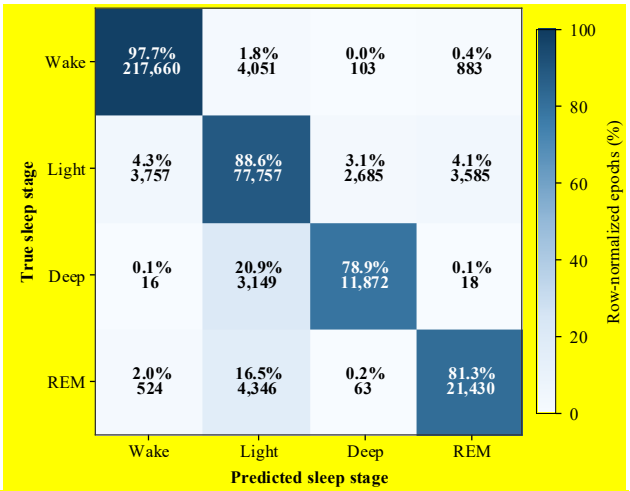


Fig. 3. Sleep-EDF confusion matrix under the Post30 wake-window strategy.

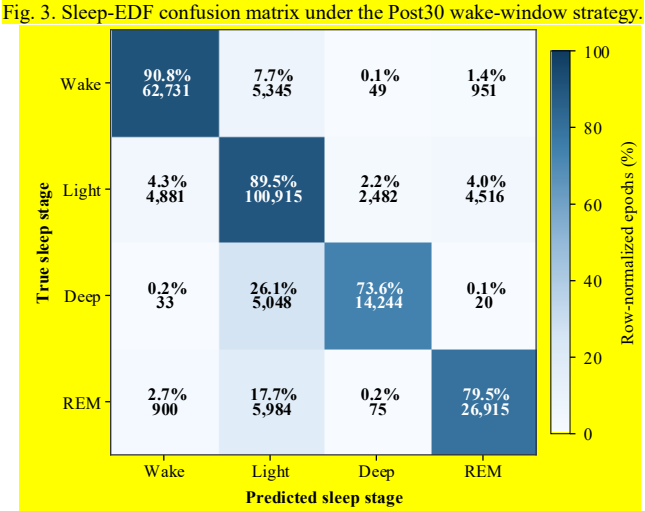


Fig. 4. Sleep-EDF confusion matrix under the trimmed wake-window strategy.

B. Model classification results of IMU data

The transferred target-domain labels were first inspected at the sequence level. A representative EEG-derived label sequence is shown in Fig. 5, where Wake, Light Sleep, Deep Sleep, and REM labels follow the expected overnight macrostructure. This visualization provides a qualitative check that the transferred labels preserve sleep-cycle continuity, rather than behaving as isolated epoch predictions. For quantitative evaluation, accuracy, macro-F1, recall, and Cohen's kappa were used with the confusion matrices.

$$\begin{cases} \mathrm{Acc} = \dfrac{\sum_{c=1}^{K} TP_c}{N}, \\ \mathrm{MF1} = \dfrac{1}{K}\sum_{c=1}^{K} \dfrac{2P_cR_c}{P_c+R_c}, \\ \mathrm{Recall} = \dfrac{1}{K}\sum_{c=1}^{K} R_c, \\ \kappa = \dfrac{p_o - p_e}{1-p_e}. \end{cases} \tag{5}$$

where TP_c denotes the number of true positives for sleep stage c, K is the number of sleep stages, and N is the total number of evaluated epochs. P_c and R_c denote the precision and

recall of stage c, respectively, where $P_c = TP_c/(TP_c+FP_c)$ and $R_c = TP_c/(TP_c+FN_c)$. In this study, K=4, corresponding to Wake, Light Sleep, Deep Sleep, and REM. For Cohen's kappa, p_o represents the observed agreement and is equivalent to Acc, while p_e denotes the expected agreement by chance and is calculated as $p_e = \frac{1}{N^2}\sum_{c=1}^{K} n_{c,\cdot}\ n_{\cdot,c}$, where $n_{c,\cdot}$ and $n_{\cdot,c}$ are the row and column totals of class c in the confusion matrix, respectively. Cohen's kappa was additionally reported to quantify agreement beyond chance, which is particularly important under imbalanced sleep-stage distributions.

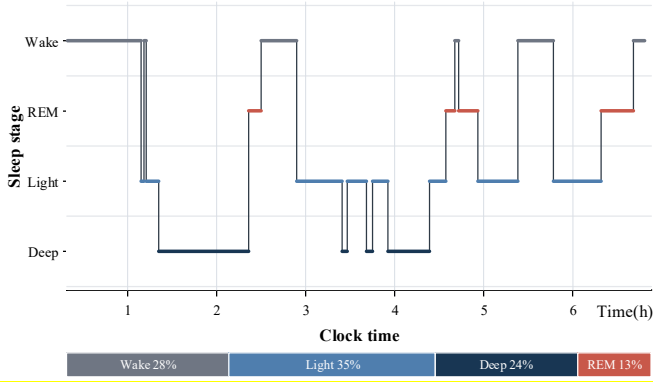


Fig. 5. The hypnogram of EEG-derived labels transferred to synchronized IMU epochs.

Table III compares the performance of different IMU-based sleep staging models under subject-wise splitting. This setting ensures that subjects in the training and test sets are mutually exclusive, thereby reducing the risk of subject-level information leakage.

TABLE III
PERFORMANCE COMPARISON OF TARGET-DOMAIN IMU-BASED SLEEP STAGING MODELS UNDER SUBJECT-WISE SPLITTING

Model	Acc	MF1	Kappa	Recall
SVM	0.7129	0.6203	0.5743	0.7249
Logistic Regression	0.5974	0.5543	0.4145	0.6937
MLP	0.6730	0.6411	0.5983	0.6391
ours	0.8356	0.7969	0.7498	0.7933

For the conventional machine-learning baselines, six-axis time- and frequency-domain features were extracted from the triaxial acceleration and triaxial angular-velocity signals following the feature construction strategy used in our previous belt-based sleep monitoring study [45]. Compared with conventional machine-learning baselines, including SVM, logistic regression, and MLP, the proposed model achieved the best overall performance, with an accuracy of 0.8356, a macro-F1 score of 0.7969, a Cohen's kappa of 0.7498, and a macro-averaged recall of 0.7933. In particular, the higher macro-F1 and kappa values indicate that the proposed model provides more balanced classification performance across sleep stages and stronger agreement with the transferred labels beyond chance. These results demonstrate the effectiveness of the proposed IMU-based model for target-domain non-wearable sleep staging. To clarify the target-domain sample distribution after subject-wise partitioning and confidence filtering, Table IV summarizes the number of participants and the class-wise epoch counts in the training, validation, and test sets.

TABLE IV
DISTRIBUTION OF IMU DATA SET

Split	Subjects	Wake	Light	Deep	REM
Training	28	7368	13425	2227	4579
Validation	4	1072	1886	333	661
test	8	2111	3828	641	1311

The confusion matrix indicates that the IMU-based model achieved reliable four-stage sleep classification, with an accuracy of 83.56%, a macro-F1 score of 79.69%, a Cohen's kappa of 0.7498, and a macro-averaged recall of 79.33%. Wake and Light Sleep were recognized with relatively high recall values of 85.5% and 87.0%, respectively, suggesting that the mattress-based IMU and physiological features captured discriminative movement and cardiorespiratory patterns associated with these stages. REM was also identified with a recall of 79.2%.

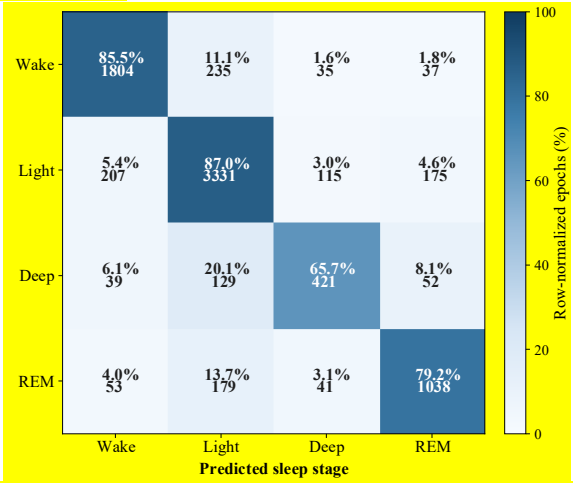


Fig. 6. Row-normalized confusion matrix of the IMU model evaluated against EEG-transferred labels.

In contrast, Deep Sleep showed the lowest recall of 65.7%, with 20.1% of Deep Sleep epochs misclassified as Light Sleep. This indicates that distinguishing NREM sleep depth remains challenging for non-EEG IMU-based signals. Most errors occurred between physiologically adjacent or behaviorally similar stages, especially Deep-to-Light and REM-to-Light transitions, which is consistent with the limited neurophysiological information available from IMU-derived signals.

IV. CONCLUSION

This paper presents a synchronized EEG-guided label-transfer framework for non-wearable sleep staging. In the source domain, the proposed sleep-staging model achieved an accuracy of 0.8662 and a Cohen's kappa of 0.7904 on Sleep-EDF, showing performance comparable to established single-channel EEG baselines. The wake-window experiments further demonstrated that apparent model performance can vary substantially, with accuracy ranging from 0.8714 to 0.9338, depending on how pre-sleep and post-sleep wake epochs are retained. In the target domain, synchronized EEG-derived pseudo-labels were transferred to temporally aligned IMU epochs collected from 40 healthy participants and used to train an IMU-based belt model. The resulting model achieved an accuracy of 83.56%, a macro-F1 score of 79.69%, and a Cohen'

s kappa of 0.7498 against the transferred labels. These findings support the feasibility of using synchronized EEG supervision to develop IMU-based sleep staging models for non-wearable deployment scenarios.

Several limitations should be acknowledged. First, the target-domain cohort comprised 40 healthy participants and did not include patients with insomnia, obstructive sleep apnea, fragmented sleep, or other sleep disorders. Therefore, the present results cannot establish model performance in clinically heterogeneous populations. Second, the four-class staging scheme merged N1 and N2 into a single Light Sleep class. This strategy improved class stability and reduced label sparsity, but it precluded separate analysis of N1-related sleep fragmentation, which may be clinically relevant. Third, the target-domain labels were generated by the EEG model rather than obtained from independent manual PSG scoring. Accordingly, the reported IMU results should be interpreted as feasibility evidence for EEG-guided label transfer and IMU-based model training, rather than as definitive evidence of clinical-grade sleep staging performance.

To address these limitations, future work will focus on two key directions. First, clinical cohorts should be recruited, including participants with moderate-to-severe obstructive sleep apnea (AHI $\geq$ 15), primary insomnia, or fragmented sleep, and the model should be evaluated using subject-wise validation with independent PSG annotations. Second, self-supervised pre-training on large-scale unlabeled IMU data should be incorporated to improve the robustness and generalizability of non-wearable sleep representations under diverse home-monitoring conditions. These steps are expected to advance the current proof-of-concept framework toward a clinically credible home sleep monitoring system while preserving the leakage-control strategies introduced in this revision.

Overall, the results from both the source and target domains demonstrate the technical feasibility of EEG-guided label transfer for non-wearable sleep staging. By leveraging EEG-derived pseudo-labels as supervisory information, the proposed framework provides a practical and scalable pathway for developing IMU-based sleep staging models under target-domain deployment conditions.

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